LearnTrace

ET617 · EDUCATIONAL APP DESIGN

A learning app that records how you learn

An interactive learning application that captures **every learner action** — clicks, page views, video actions, and quiz attempts — and stores it as Moodle-format clickstream data.

Text · Video · Quiz | Clickstream tracking | CSV export | Analytics dashboard

Sanidhya Katiyar · 23B2244

github.com/Sanidhya-Katiyar/Basic-App-Design-Assignment---ET617

THE ASSIGNMENT

What we were asked to build

- A **learning application** where users log in as **learners**
- Offering **interactive content** — text, video, quizzes
- Track **all user activity**: clicks, page views, video actions, quiz attempts
- **Store** that clickstream data
- A standalone **web app**, any tech stack, using **git**

THE REFERENCE

A real Moodle log export

The brief pointed to a Moodle clickstream export as the data model — a seven-column activity log.

```
Time · Event context · Component ·
Event name · Description · Origin · IP
```

We made this format the core of the app: LearnTrace emits and exports events in exactly this shape.

OVERVIEW

What LearnTrace does

01
Learner login
Session-cookie auth over scrypt-hashed passwords, with multiple demo learners.

02
Interactive content
A text lesson, a tracked video, an external resource, and a graded quiz.

03
Full activity tracking
Clicks, page views, video play/pause/seek/progress, and the quiz lifecycle.

04
Clickstream storage
Every action becomes one row in a SQLite `events` table.

05
Analytics dashboard
Live totals, charts, quiz performance, and an activity feed.

06
Moodle-format export
Download the log as a CSV matching the exact seven Moodle columns.

UNDER THE HOOD

Tech stack

FRONTEND

React · TypeScript · Vite

A single-page app with client-side routing and a hand-built CSS design system.

BACKEND

Node.js · Express

REST API for auth, content, quiz grading, event ingestion, analytics, and export.

DATABASE

SQLite · node:sqlite

Node's built-in driver — zero native builds, a single file, fully local.

WHY THIS STACK

Runs offline, one command

No cloud keys, ideal for a demo, and every layer is easy to inspect.

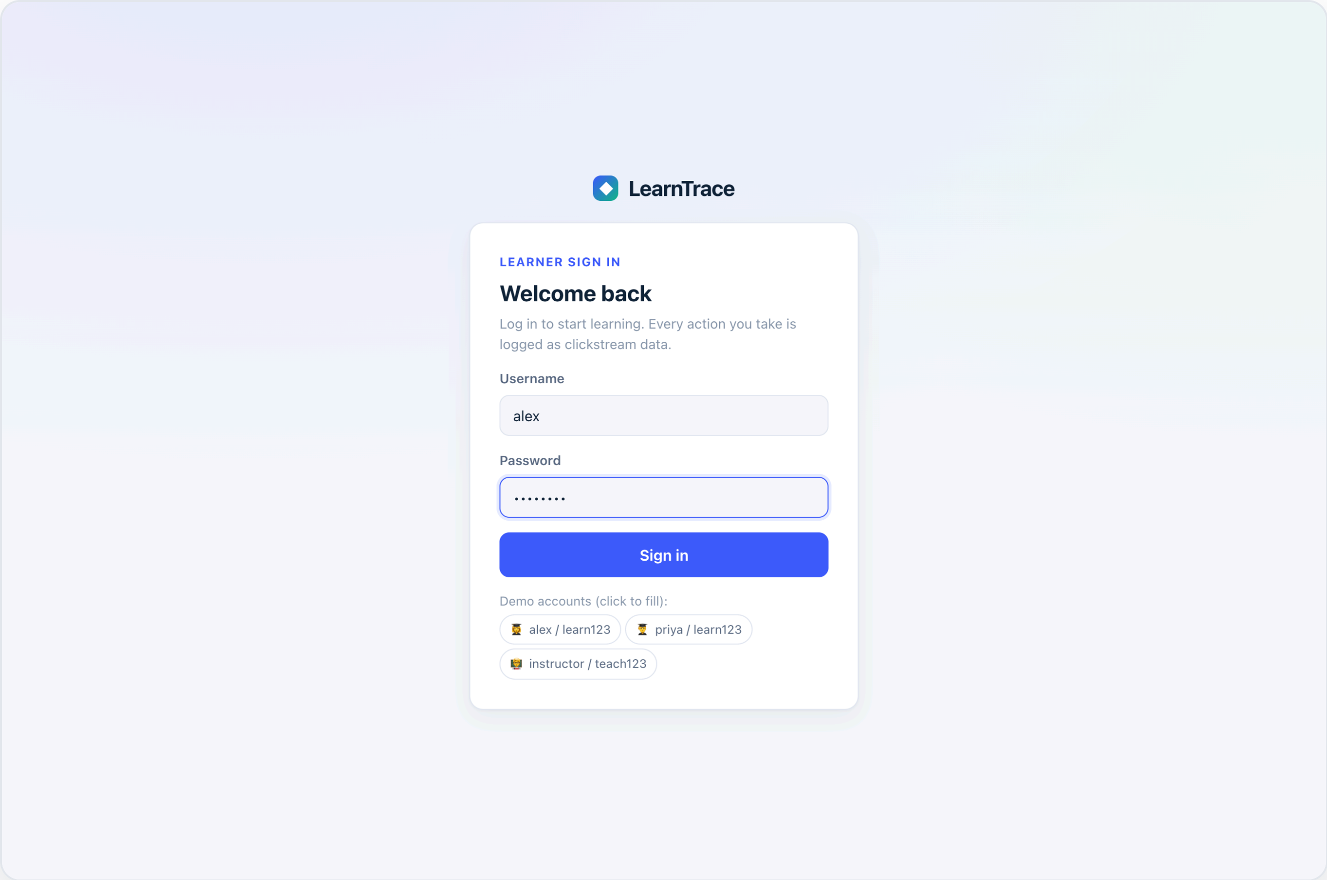
React | TypeScript | Vite | Express | node:sqlite | scrypt auth | sendBeacon | git

WALKTHROUGH · 1

Learners log in

A clean sign-in with demo accounts. The **login event is recorded server-side** with the session, IP, and user agent — the first row in the clickstream.

```
alex / learn123 · learner
priya / learn123 · learner
instructor / teach123
```

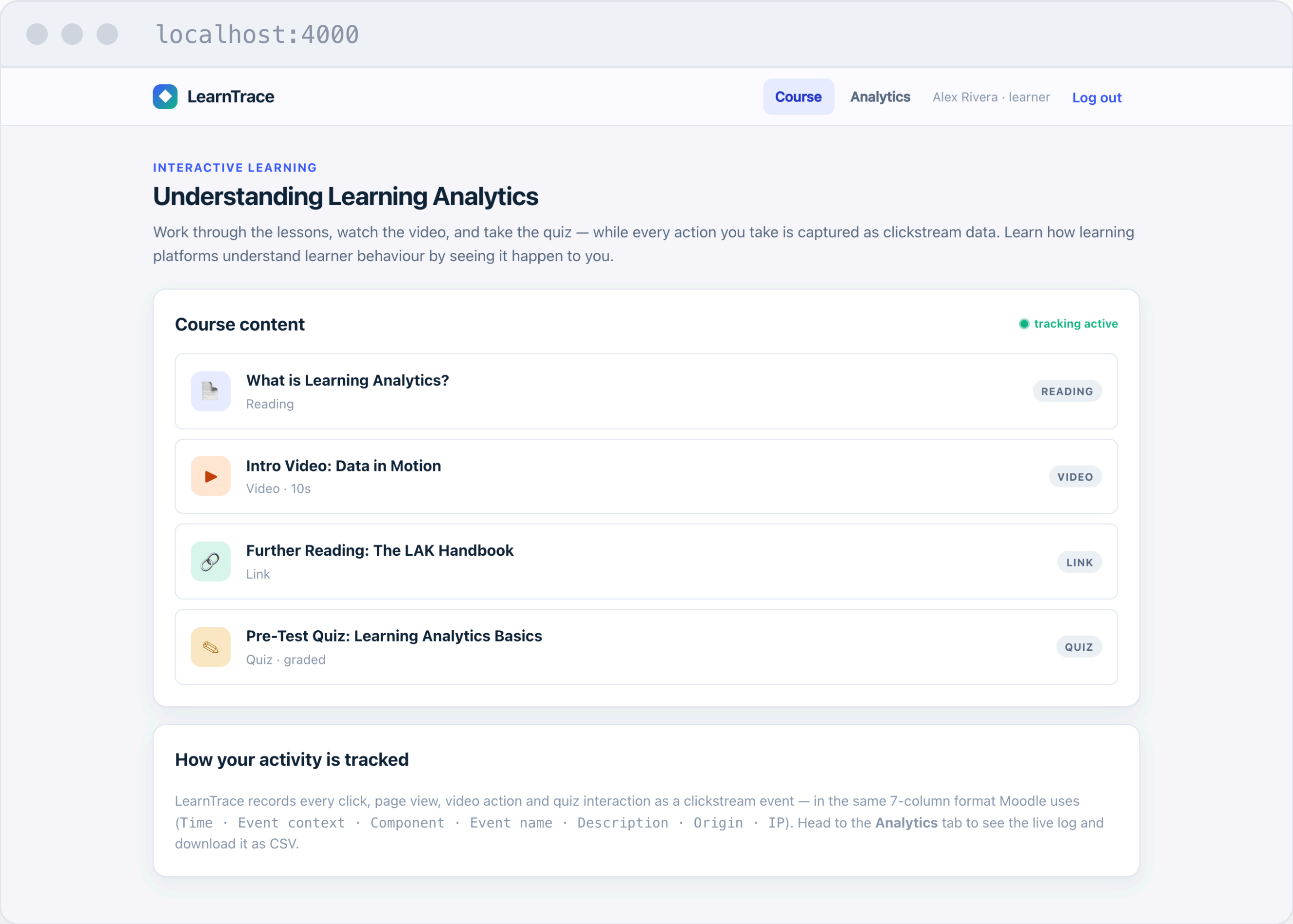


WALKTHROUGH · 2

Interactive content

One course with **four activity types** — reading, video, an external link, and a graded quiz. Opening any of them logs a `Course module viewed` event.

Text | Video | URL | Quiz

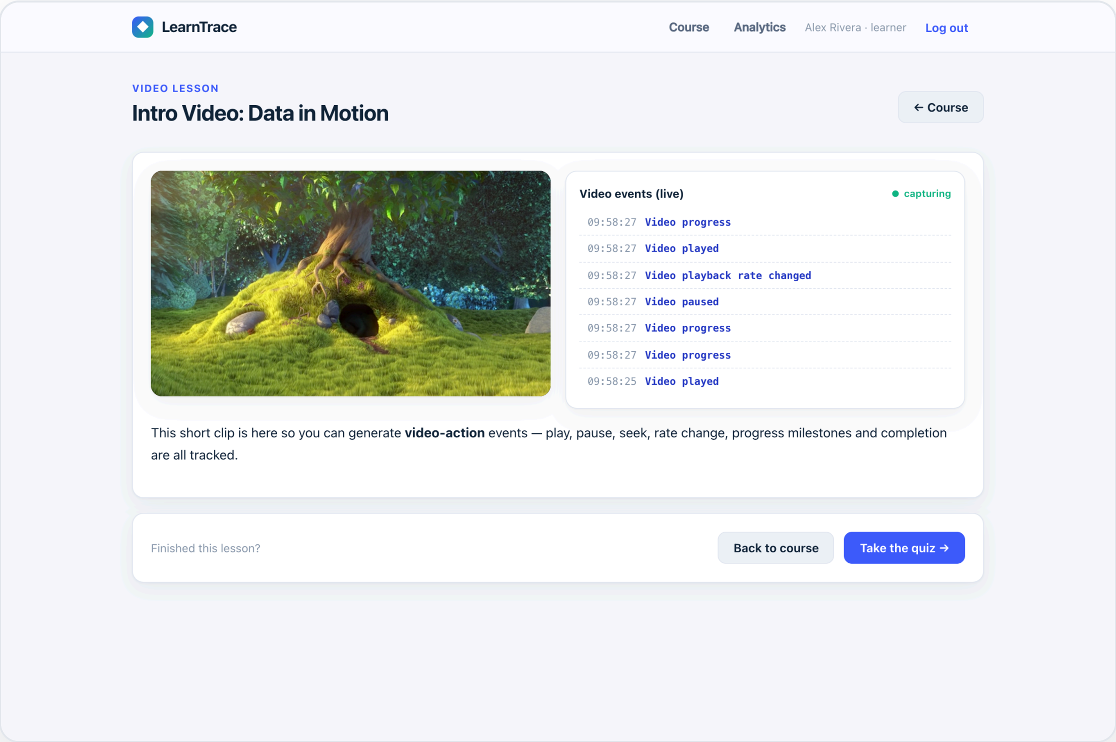


WALKTHROUGH · 3

Video actions, tracked live

The HTML5 player emits an event for every interaction — and a **live feed** shows them landing in real time.

- play · pause · **seek** · rate change
- progress milestones (25 / 50 / 75%)
- completion · fullscreen



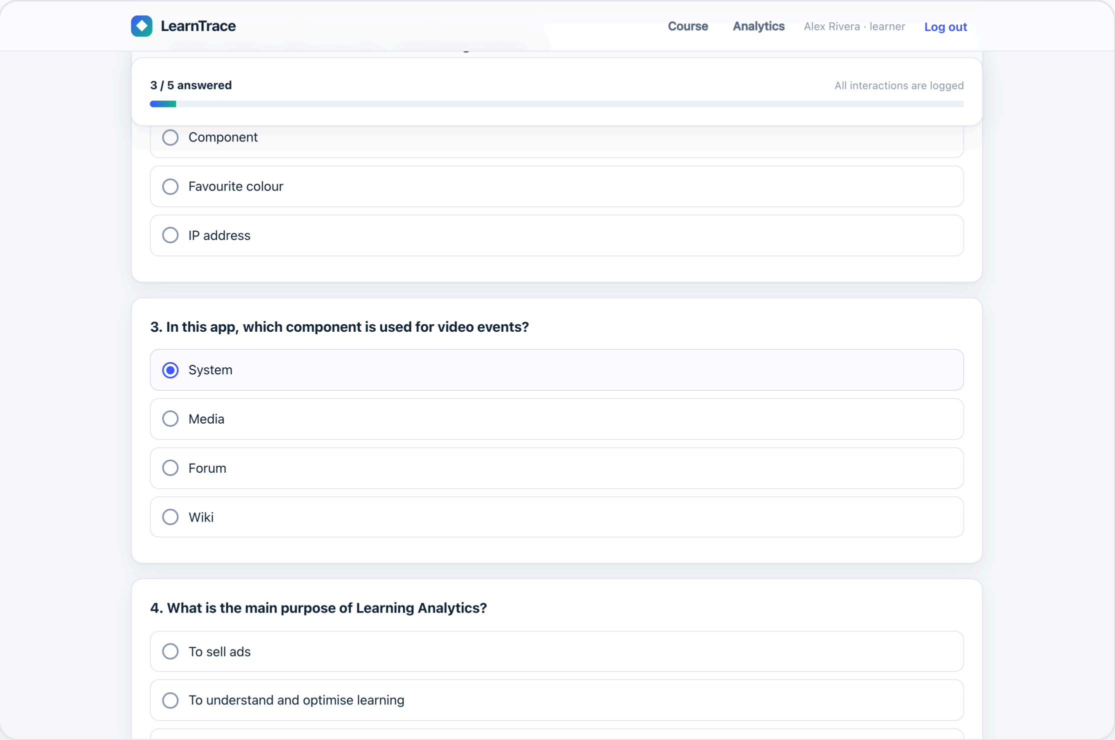
WALKTHROUGH · 4

The full quiz lifecycle

Every stage is a distinct event, so we can reconstruct exactly how a learner worked through the quiz.

- `Quiz attempt started`
- `Quiz answer changed` — per selection
- `Quiz attempt submitted` → `User graded`

Grading is **server-side** — answers are never exposed to the client.



THE CORE IDEA

How the clickstream is captured

CLIENT

Capture

Delegated click listener, route-change page views, and video/quiz hooks queue events.

→

CLIENT

Buffer & flush

Batched every 5s, on route change, and on unload via `sendBeacon`.

→

SERVER

Enrich

Stamps time, user, IP, and a Moodle-style sentence.

→

DATABASE

Store

One row per event in the SQLite `events` table.

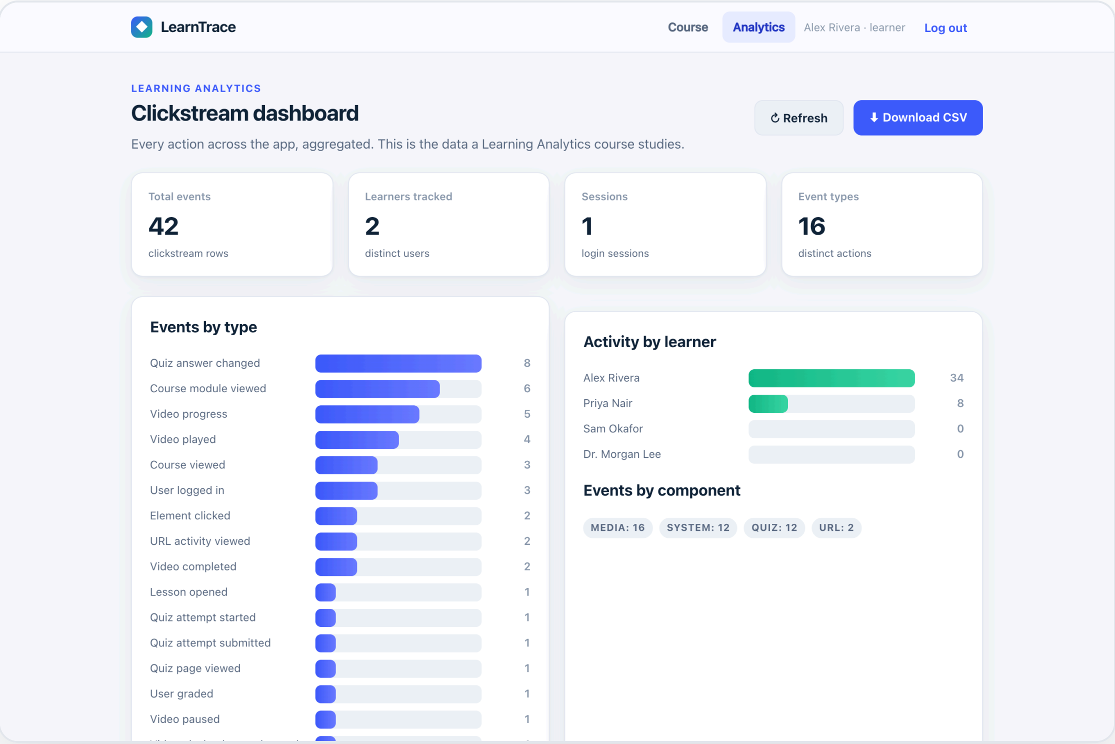
Why sendBeacon? In a single-page app, navigating away would normally drop buffered events. `navigator.sendBeacon` guarantees the final batch is delivered even as the page unloads.

WALKTHROUGH · 5

Analytics dashboard

The stored clickstream, made legible — the data a learning-analytics course actually studies.

- Events by type and by component
- Activity per learner
- Quiz performance and a live feed



THE PAYOFF

Exports in exact Moodle format

One click downloads the whole clickstream as a CSV whose columns match the instructor's Moodle sample **exactly** — the same shape real learning platforms produce.

Time	Event context	Component	Event name	Description	Origin	IP address
28/07/26, 14:31:04	Course: ET617 Educational…	System	User logged in	The user with id '1' logged in.	web	10.0.0.1
28/07/26, 14:35:19	Video: Intro Video	Media	Video played	…played the video 'Intro' at 00:0…	web	10.0.0.1
28/07/26, 14:38:02	Quiz: Pre-Test Quiz	Quiz	Quiz attempt submit…	…submitted the attempt scoring 6/…	web	10.0.0.1

BY THE NUMBERS

One demo session captured

45 clickstream events	**17** distinct event types	**4** components tracked	**7** Moodle CSV columns

Components tracked: System Media Quiz URL

LearnTrace

DELIVERABLES

Built, documented, shipped

- ✓ Working web app — login, content, tracking, dashboard
- ✓ GitHub repository with README
- ✓ Clickstream stored and CSV export
- ▶ Demo video and this presentation

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Thank you